

Practice Problems -7

A thin lens with an aperture of 4.80 cm and a focal length of +3.50 cm has a 3.0-cm stop located 1.50 cm in front of it. An object 1.50 cm high is located with its lower end on the axis 8.0 cm in front of the lens. Locate graphically and by formula (a) the position and (b) the size of the exit pupil. (c) Locate the image of the object graphically by drawing the two marginal rays and the chief ray from the top end of the object.

A thin lens with a focal length of +6.0 cm has an aperture of 6.0 cm. A 6.0-cm stop is located 2.0 cm in front of the lens, and a 4.0-cm stop is located 2.0 cm behind the lens. An object 4.0 cm high is centrally located on the axis 12.0 cm in front of the lens. Find the images of the two stops, and determine (a) the stop of the system, (b) its size, and (c) its position with respect to the lens. (d) Locate the image and determine its size by drawing the two marginal rays and the chief ray from the top end of the object. (e) Solve graphically (Fig. P7.4).

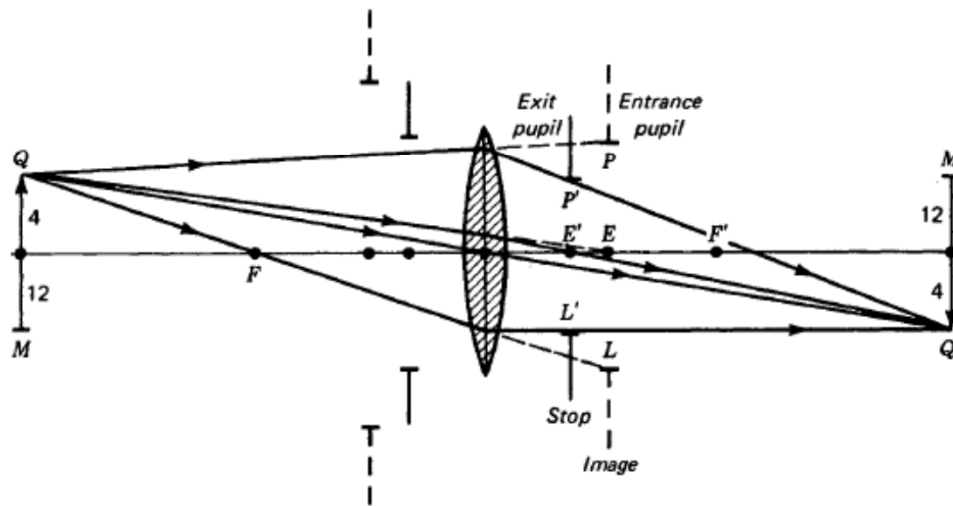


FIGURE P7.4
Graphical solution for Prob. 7.4.